

Structure-Based Protein Function Prediction via Graph Convolutional Networks

Presentation of BMEG3102 – Bioinformatics

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Outline

1 Introduction: Protein Function Prediction

2 Previous Works

3 Modelling DeepFRI

Network Architecture

Experimental Methods

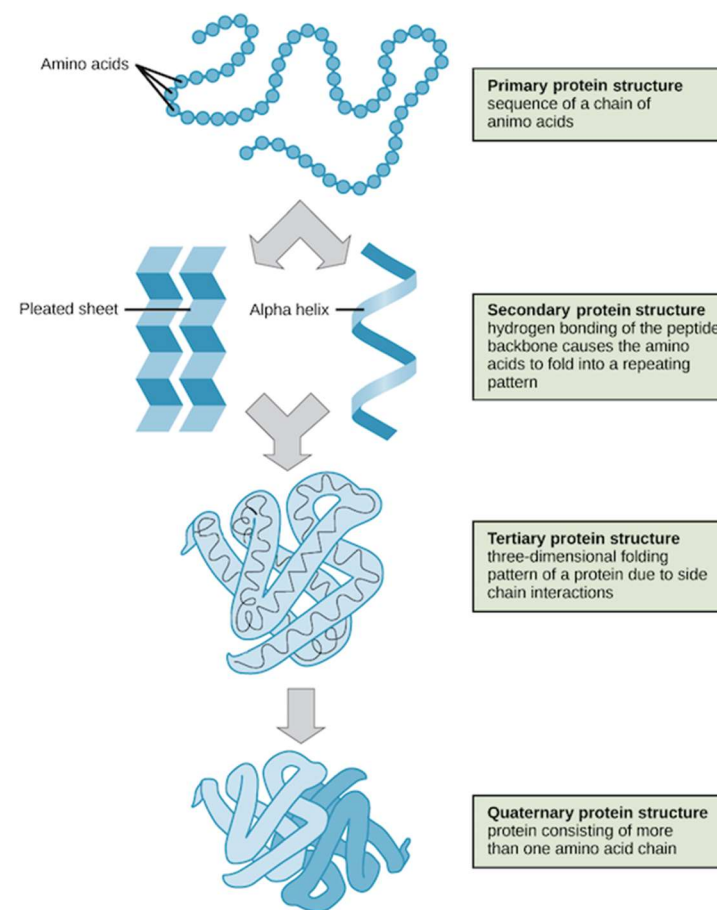
Quantitative Justifications

Qualitative Justifications

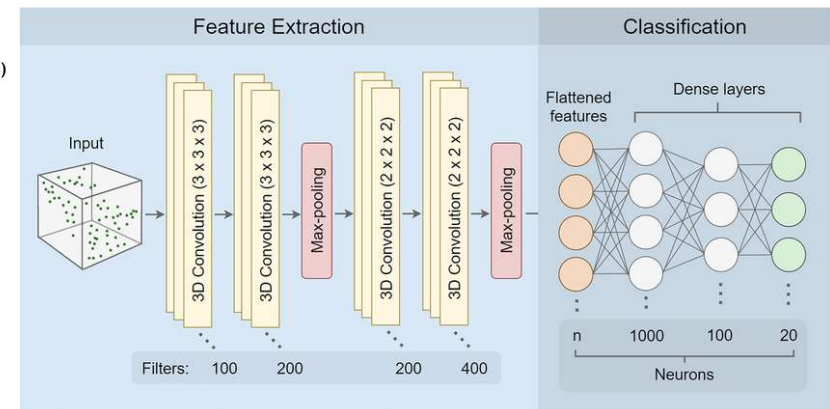
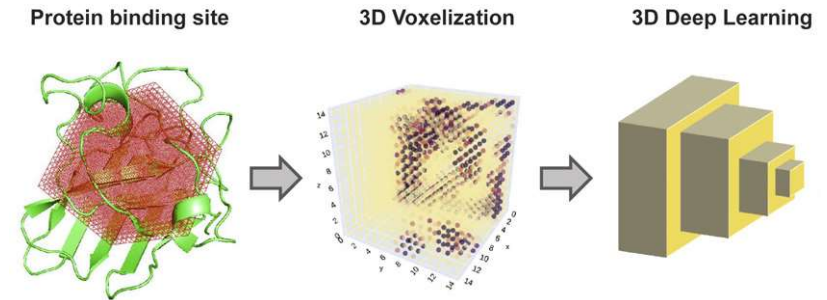
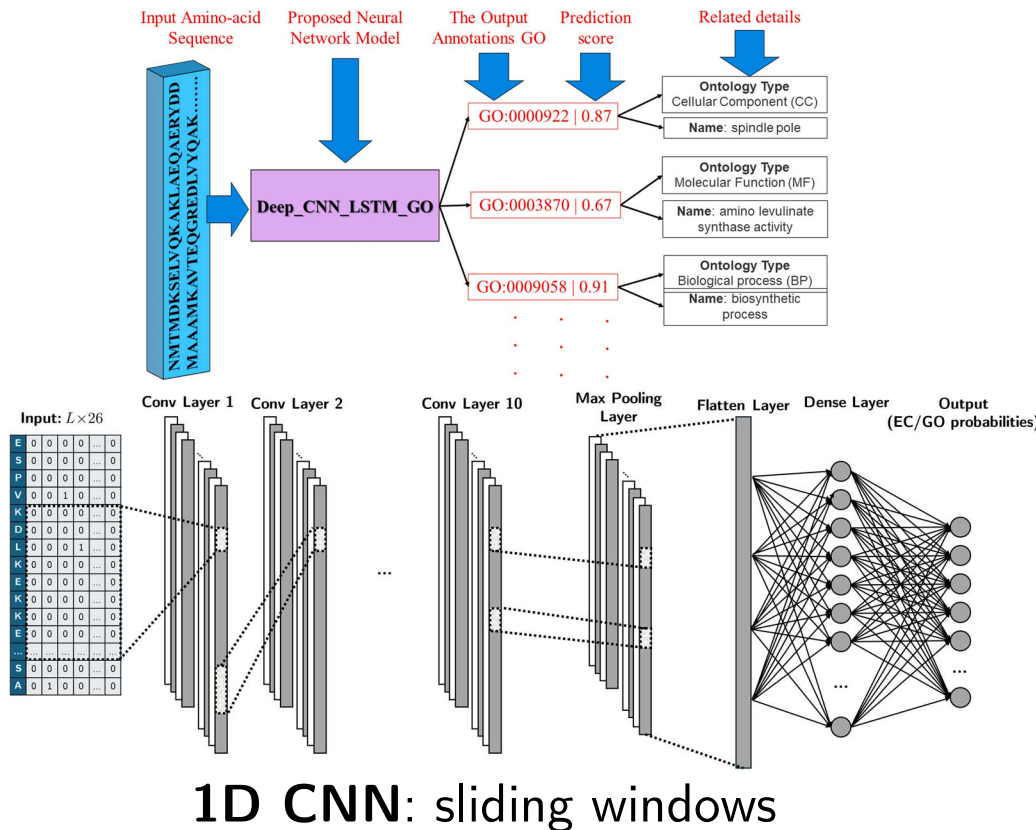
4 Summary

Problem Description

- ▶ **Protein: 3D structured**, with a wide range of biological functions.
- ▶ Biological functions are annotated with **Gene Ontology (GO) terms** or **Enzyme Commission (EC) numbers**.
- ▶ Experimental approaches: large-scale, costly, requires careful design
- ▶ **Goal:** Study protein functions via computational biology.
- ▶ Handcrafted features are prone to be imprecise and erroneous.
- ▶ **Alternatives: Machine Learning Approaches**



Previous Works



3D CNN: spatial localized filters

- 1D CNN: What if the **folding** is complex?
- 3D CNN: Space is always **sparse**, precision is crucial.

Fails to consider the **tertiary** structure

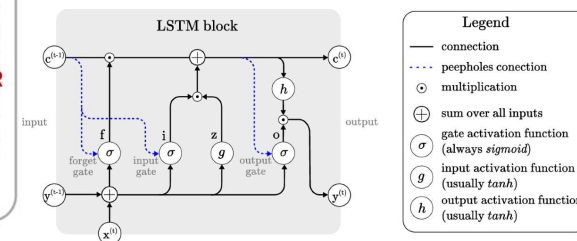
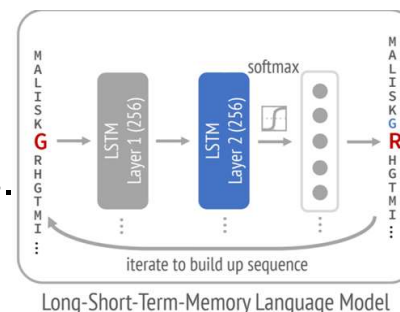
Imposes constraints on **protein length and complexity**

Question: Is there a better way to model spatial features, robust to structural errors?

Proposed Network Architecture – DeepFRI

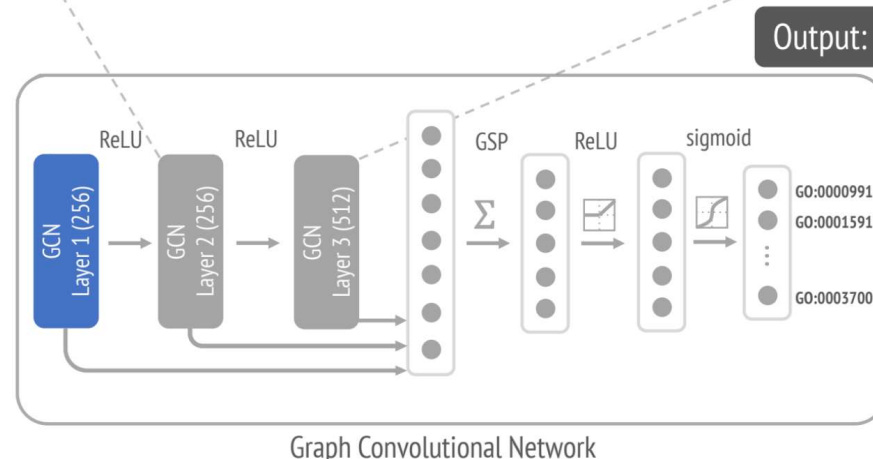
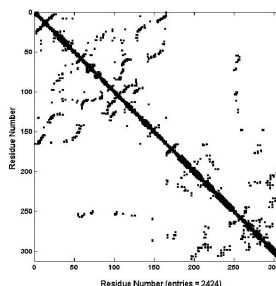
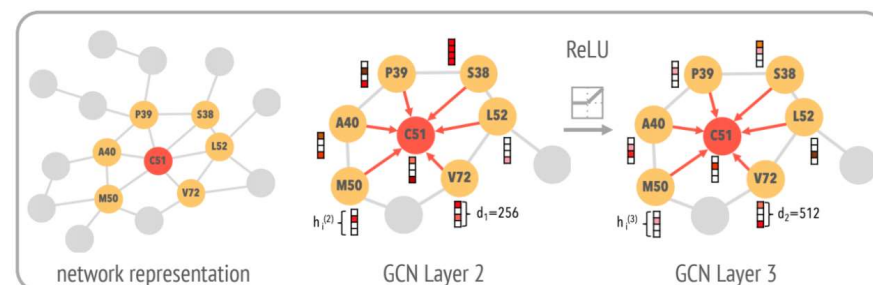
First stage: LSTM Language Model

- Purpose: extracting **residue-level features (motifs and domains)** from PDB sequences.
- Input: amino acid sequence
- Output: 2nd LSTM layer (encode node features)
- Trained with Pfam to **predict an amino acid residue** based on its context and position.

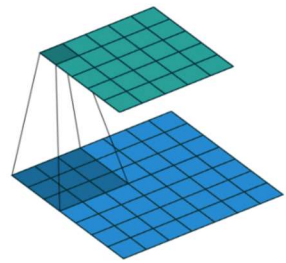


Second stage: Graph Convolutional Network

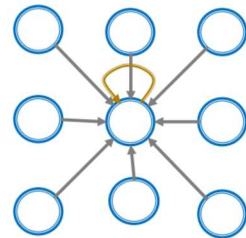
- Purpose: convolve the residue-level features **proximal** in **structure**
- Input: features from LSTM-LM, contact map
- Output: function probabilities



Sidenote: From CNN to GNN



(a) Image

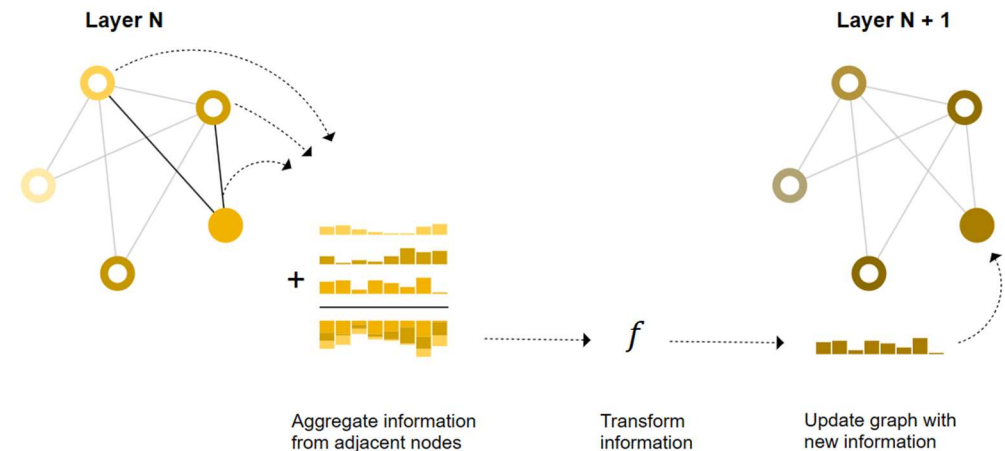
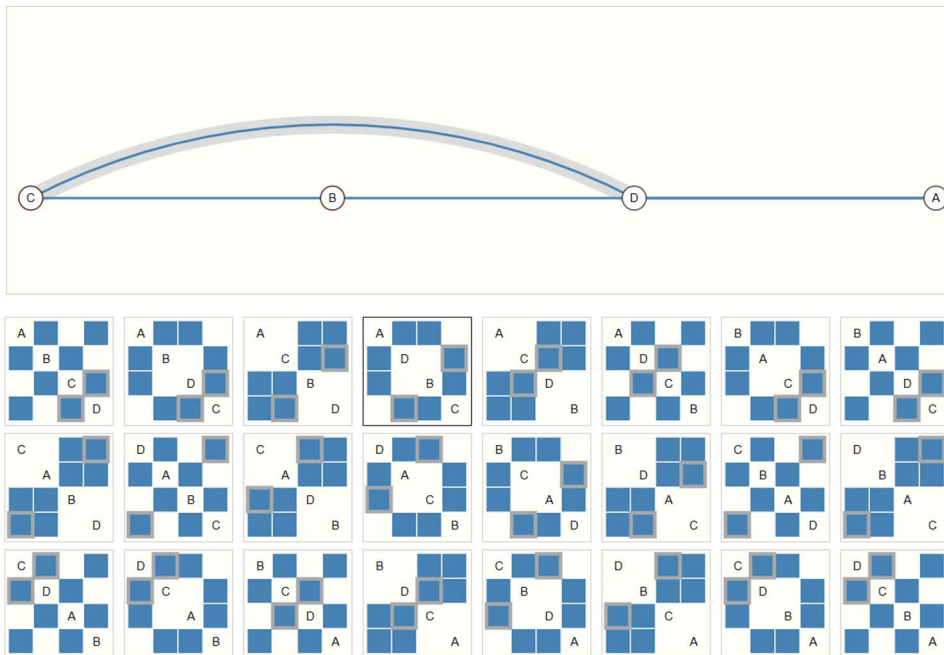


(b) Graph

Direct use of CNN is unwise.

Three distinct properties:

- (i) Isomorphism-preserving translation
- (ii) Permutation invariance
- (iii) Multi-level localization



Experimental Methods

1. Obtaining Annotated Training Samples

Protein Data Bank (PDB)

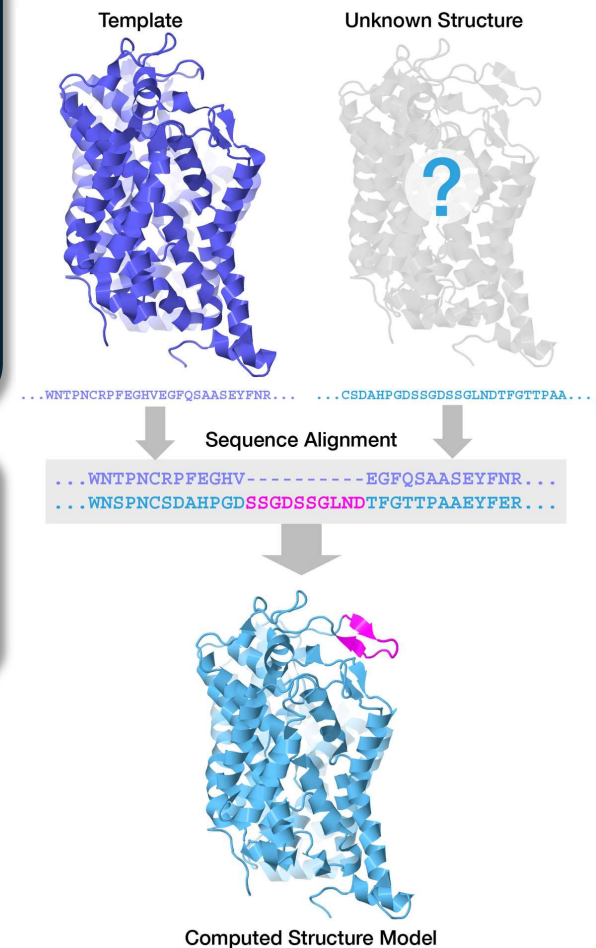
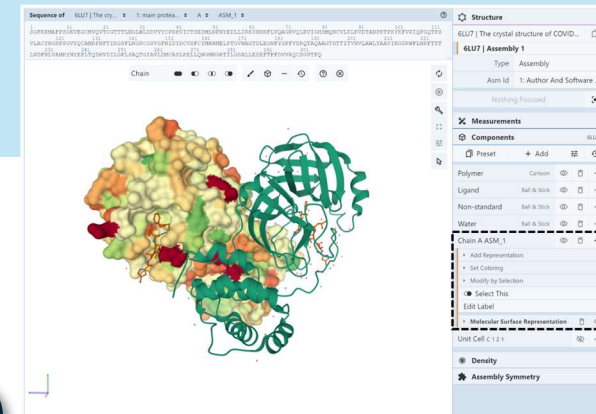
- Experimentally determined
- High quality

SWISS-MODEL

- More diversified (annotated, but no structure available)
- Homology modelling: reconstruction by a related “template”

2. Data Preparation

- Remove redundant sequences by clustering.
- Construct contact maps.



Experimental Methods

3. Evaluation Metrics

Follows standards of [The Critical Assessment of Function Annotation \(CAFA\)](#).

(1) Protein-centric maximum F-score ($F_{\max}$)
 “Peak” performance of assigning GO terms/EC numbers to a protein, as a harmonic mean of the precision and recall:

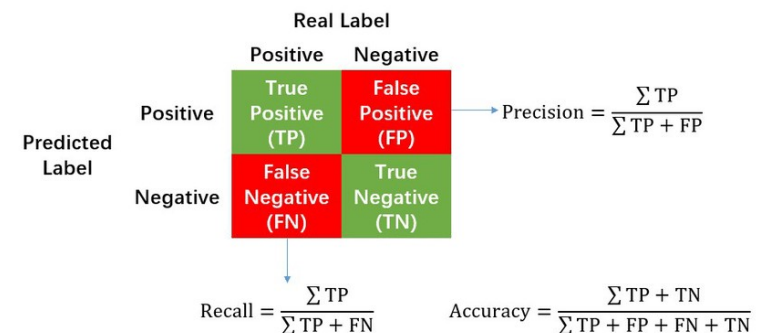
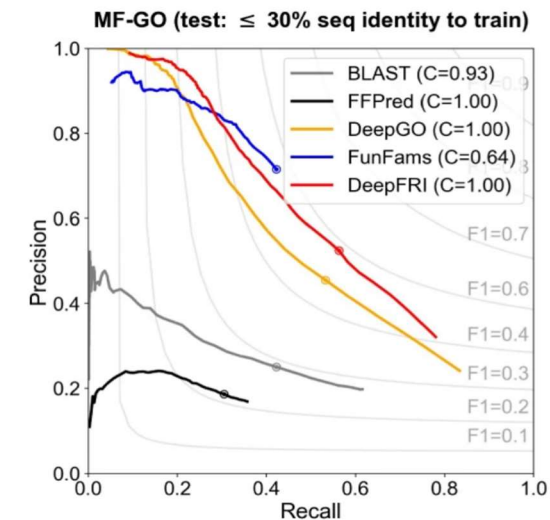
$$F_{\max} = \max_{\tau} F_1(\tau), \text{ where}$$

$$F_1(\tau) = 2 \times \frac{\text{Precision}(\tau) \times \text{Recall}(\tau)}{\text{Precision}(\tau) + \text{Recall}(\tau)}$$

(2) Term-centric area under precision-recall (AUPR) curve

Measures the model’s ability to balance precision and recall:

$$\text{AUPR} = \int_0^1 \text{Precision}(\tau) d\tau$$

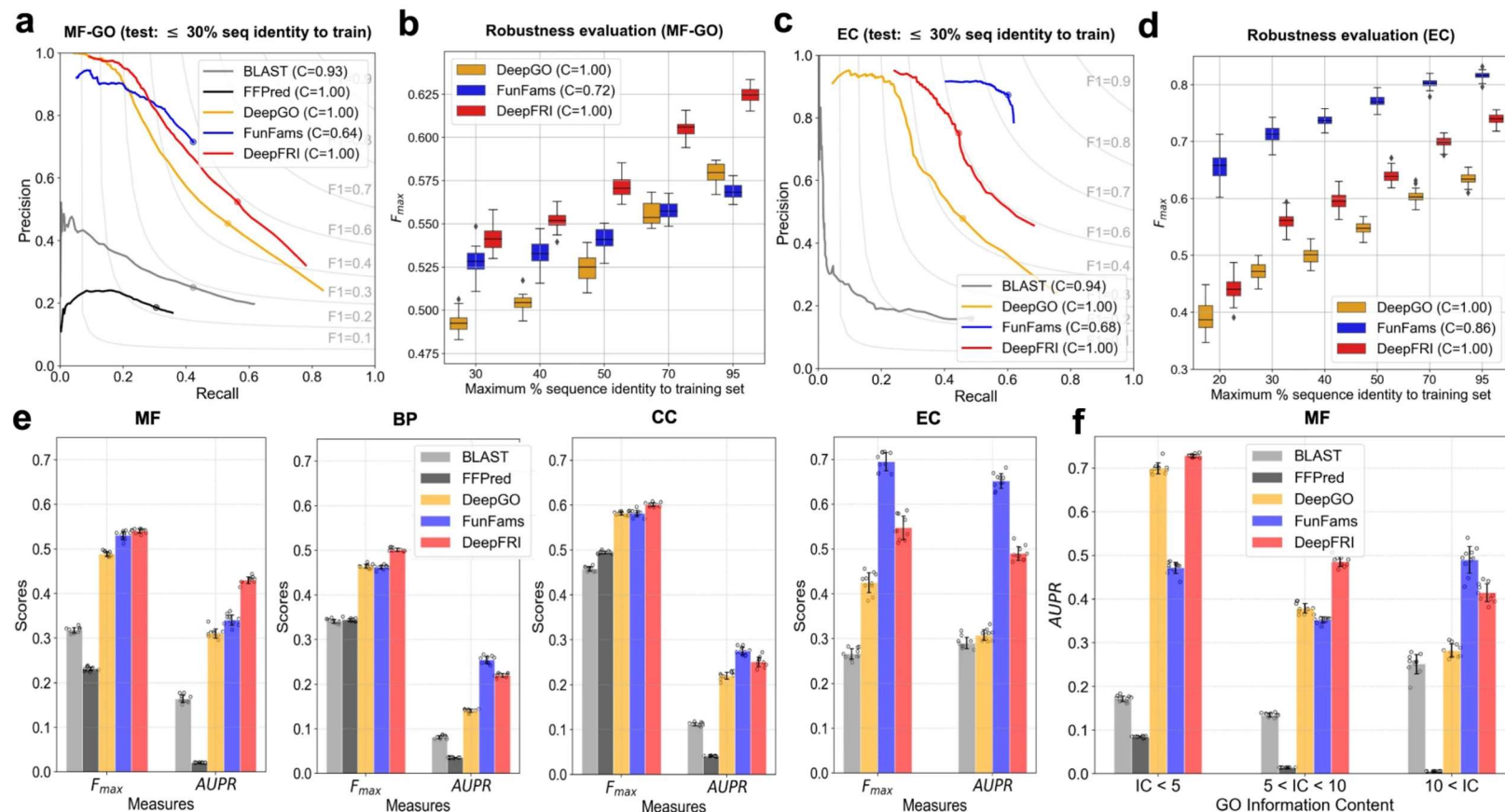


Larger F-score and larger area under the AUPR curve implies better performance of the model.

Quantitative Experiment 1

Question: How good is DeepFRI when compared to DeepGo, BLAST, and FunFams?

- Investigate the performance of DeepFRI trained on precise PDB structures.

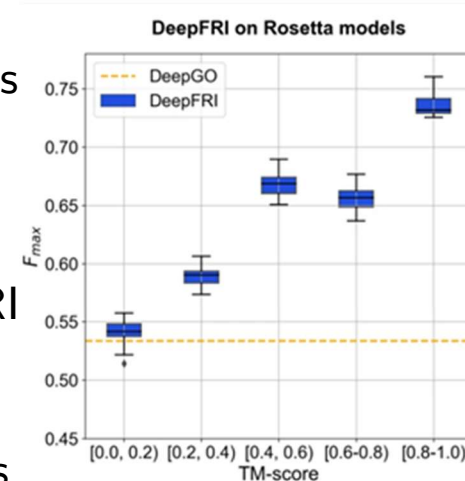
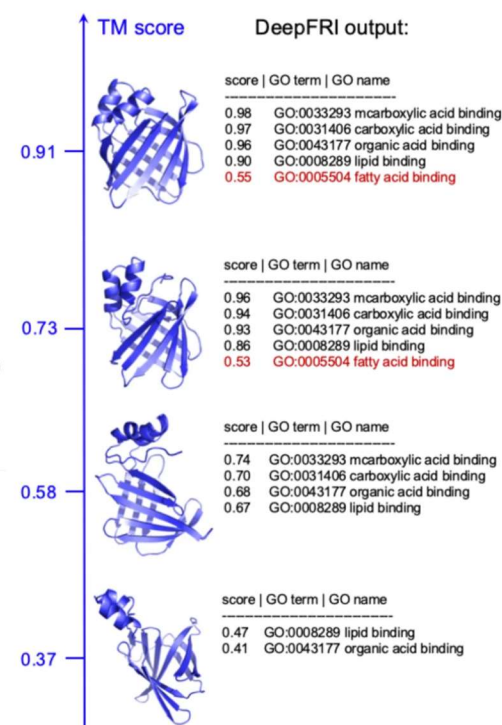
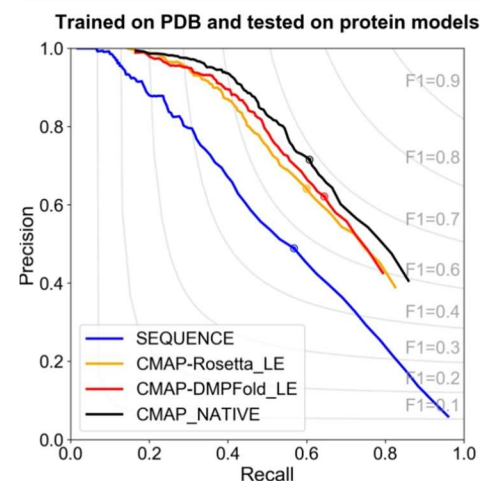


DeepFRI outperforms other methods on MF- and BP-GO terms (Fig. 2a,e). DeepFRI also learns general structure-function relationships more robustly than other methods by predicting MF-GO terms of proteins with low sequence identity to the training set (Fig. 2b).

Qualitative Experiment 2

Question: Is DeepFRI sensitive to the modeling errors in protein structure prediction?

- ▶ Construct contact maps on a test set with 700 unseen PDB chains.
- ▶ Build s with NATIVE structures, lowest energy model by Rosetta and DMPFold. Compare the $F_{\max}$ score with the CNN-based DeepGO
- ▶ Further create different template modelling scores (TM-scores) Rosetta models against the NATIVE structures.

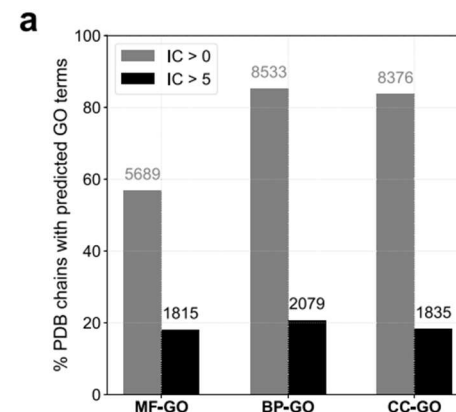
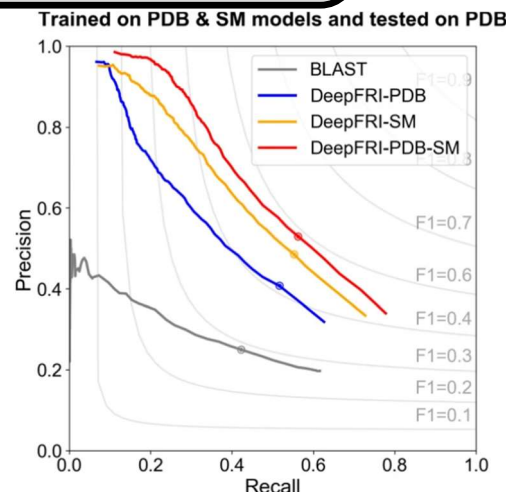


1. DeepFRI exhibits higher performance (with $F_{\max} = 0.657/0.633/0.619$ for native structures, models from DMPFold, Rosetta, respectively).
2. For models with TM-scores > 0.58 , DeepFRI correctly predicts four GO terms including lipid binding, whereas for a TM-score > 0.73 , DeepFRI correctly predicts even more specific function - fatty acid binding.
3. **Language model** tolerates fuzzy, noisy structures.

Qualitative Experiment 3/4

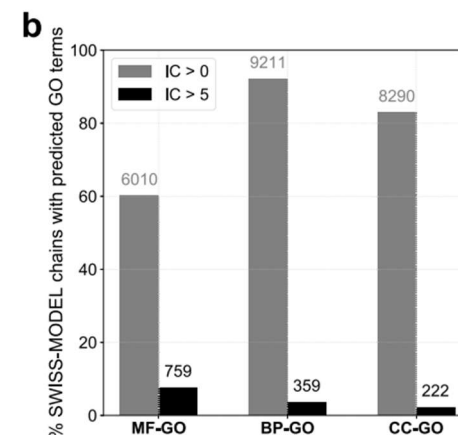
Question: How to tackle the imbalancedness in the number of positive and negative examples?

- ▶ More data! Data sanity and model generalization ability trade-offs...
- ▶ Relying on PDB is not enough anymore.
- ▶ SWISS-MODEL: ~ 220k nonredundant homology models.



Question: How to evaluate DeepFRI's performance in a realistic scenario?

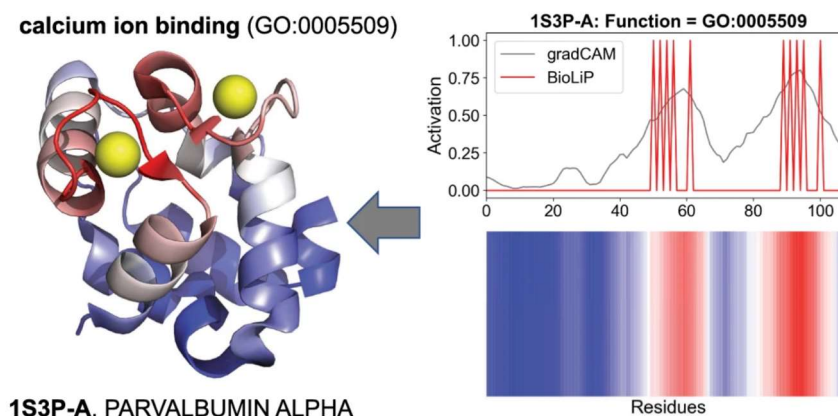
- ▶ Temporal holdout: Retain some data for testing.
- ▶ Check if BLAST and DeepGO generates any reasonable result.



Qualitative Experiment 5

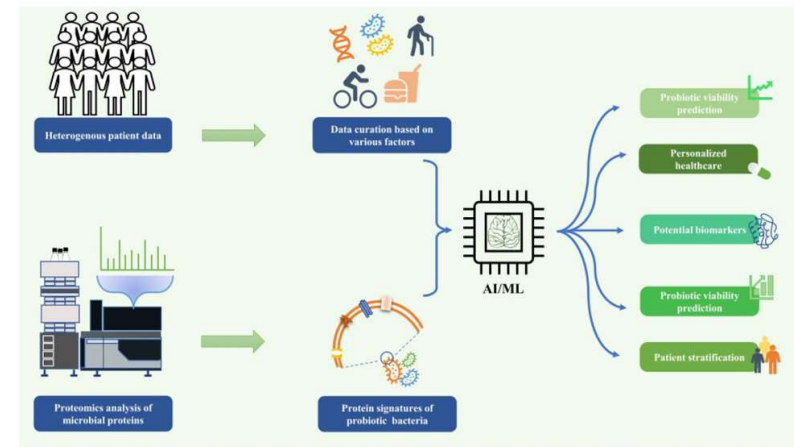
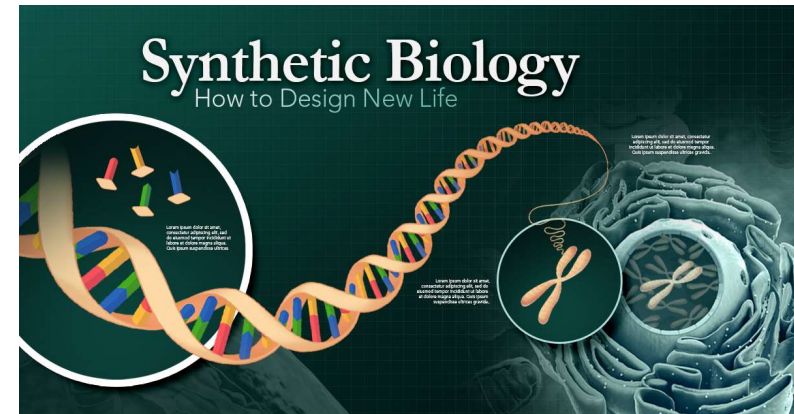
Question: How to further justify the robustness of DeepFRI in terms of its ability to leverage structural information?

- ▶ Check where a protein chain has its bindings annotated using post-processing techniques.
- ▶ Gradient-weighted Class Activation Maps (grad-CAM)
 1. Compute the **contribution** of each graph convolutional feature map
 2. Sum the feature maps with positive contribution
- ▶ High area under ROC curve indicates high correspondence between annotated binding sites and predictions.



Conclusion

- ▶ Better modeling of structures using **LSTM-LM and GNN** makes DeepFRI a SOTA method.
- ▶ Lesson learned: considerate **experimental designs** provides means of interpreting prediction results.
- ▶ **Future works**: data coverage on proteins with rare GO terms/EC numbers, improvements on model architecture, new models covering the quaternary structure.
- ▶ DeepFRI and future machine-learning-driven ML approaches will continue to pave the way for breakthroughs in drug discovery, synthetic biology, and molecular medicine.



Any Questions?

Thanks!

